All-Orders Core-Hole Potential

Starting point

The overlap between the final relaxed orbital ϕ_p and the initial orbital ψ_q is approximated in first-order perturbation theory as

$$S_{pq} = \langle \phi_p | \psi_q \rangle \approx \frac{\langle \psi_p | \hat{v}_{\rm ch} | \psi_q \rangle}{\epsilon_p - \epsilon_q}, \qquad p \neq q. \tag{1}$$

Here we show that this is the first term of a series whose sum is obtained by diagonalising a matrix, and we state the model this rests on.

The perturbation $\hat{V} = -\hat{J}_c$

Removing one electron from the core orbital ψ_c removes its Coulomb field, $v_{ch}(\mathbf{r}) = -\int d\mathbf{r}'\, \rho_c(\mathbf{r}')/|\mathbf{r} - \mathbf{r}'|$, with $\rho_c = |\psi_c|^2$. This is minus the Coulomb operator of ψ_c,

$$\hat{V} \equiv \hat{v}_{\rm ch} = -\hat{J}_c, \qquad \hat{J}_c\, \psi(\mathbf{r}) = \int d\mathbf{r}'\, \frac{\psi_c^*(\mathbf{r}')\psi_c(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\, \psi(\mathbf{r}). \tag{2}$$

Expanding the orbitals in the AO basis, $\psi_p(\mathbf{r}) = \sum_\alpha C_{\alpha p}\chi_\alpha(\mathbf{r})$ and $\psi_c(\mathbf{r}) = \sum_\gamma C_{\gamma c}\chi_\gamma(\mathbf{r})$, its AO matrix is a contraction of the two-electron integrals with the core density,

$$V_{\alpha\beta} = -(J_c)_{\alpha\beta} = -\sum_{\gamma\delta} C^*_{\gamma c} C_{\delta c}\, (\alpha\beta|\gamma\delta), \qquad (\alpha\beta|\gamma\delta) = \iint d\mathbf{r}\, d\mathbf{r}'\, \frac{\chi^*_\alpha(\mathbf{r})\chi_\beta(\mathbf{r})\chi^*_\gamma(\mathbf{r}')\chi_\delta(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \tag{3}$$

Transforming to the initial MO basis gives the four-centre MO integral,

$$V_{pq} = \langle \psi_p | \hat{V} | \psi_q \rangle = \sum_{\alpha\beta} C^*_{\alpha p} V_{\alpha\beta} C_{\beta q} = -\sum_{\alpha\beta}\sum_{\gamma\delta} C^*_{\alpha p} C_{\beta q} C^*_{\gamma c} C_{\delta c}\, (\alpha\beta|\gamma\delta) = -(pq|cc). \tag{4}$$

Model and perturbation series

Assumption: the initial orbitals are eigenfunctions of an unperturbed one-electron operator $\hat{f}_0$ with eigenvalues ϵ_p, i.e. $\hat{f}_0 = \mathrm{diag}(\epsilon)$ in the $\{\psi_p\}$ basis. This is already implicit in Eq. (1). The relaxed orbitals are taken as the eigenfunctions of $\hat{f}_0 + \lambda\hat{V}$,

$$(\hat{f}_0 + \lambda\hat{V})\, \phi_p(\lambda) = E_p(\lambda)\, \phi_p(\lambda), \qquad \lambda = 1 \text{ for the physical core hole.} \tag{5}$$

For a non-degenerate level, $\phi_p(\lambda)$ and $E_p(\lambda)$ are analytic near $\lambda = 0$, so $\phi_p = \sum_k \lambda^k \phi_p^{(k)}$ and $E_p = \sum_k \lambda^k E_p^{(k)}$. Writing $c_{qp}^{(k)} = \langle \psi_q | \phi_p^{(k)} \rangle$ with $c_{qp}^{(0)} = \delta_{qp}$ and $c_{pp}^{(k)} = 0$ for $k \geq 1$ (intermediate normalisation), and collecting powers of λ, gives the Rayleigh–Schrödinger recursion

$$(\epsilon_p - \epsilon_q)\, c_{qp}^{(k)} = \sum_r V_{qr}\, c_{rp}^{(k-1)} - \sum_{j=1}^{k-1} E_p^{(j)}\, c_{qp}^{(k-j)}, \qquad E_p^{(k)} = \sum_r V_{pr}\, c_{rp}^{(k-1)}. \tag{6}$$

First order ($k = 1$) recovers Eq. (1): $c_{qp}^{(1)} = V_{qp}/(\epsilon_p - \epsilon_q)$, so

$$S_{pq} = \langle \phi_p | \psi_q \rangle \approx \frac{V_{pq}}{\epsilon_p - \epsilon_q} = \frac{(pq|cc)}{\epsilon_q - \epsilon_p}. \tag{7}$$

Second order adds paths through a third orbital r and the level shift of the hole, $V_{pp} = -(pp|cc)$:

$$c_{qp}^{(2)} = \frac{1}{\epsilon_p - \epsilon_q}\left[\sum_{r\neq p,q} \frac{V_{qr}V_{rp}}{\epsilon_p - \epsilon_r} + \frac{(V_{qq} - V_{pp})\, V_{qp}}{\epsilon_p - \epsilon_q}\right]. \tag{8}$$

The second term is the first correction from expanding $V_{qp}/[(\epsilon_p + V_{pp}) - (\epsilon_q + V_{qq})]$: the hole moves the levels, and first order ignores this.

All orders. The series is the Taylor expansion of the exact eigenvectors of $\hat{f}_0 + \lambda\hat{V}$ in λ. At $\lambda = 1$ they follow from diagonalising

$$\boxed{[\mathrm{diag}(\epsilon) - \mathbf{J}_c]\,\mathbf{U} = \mathbf{U}\,\mathrm{diag}(E), \qquad (\mathbf{J}_c)_{pq} = (pq|cc), \qquad \phi_p = \sum_q \psi_q U_{qp} \;\Rightarrow\; S_{pq} = U_{qp}.} \tag{9}$$

Each eigenvector is labelled by its largest overlap with an initial orbital, and its phase is chosen so that $S_{pp} > 0$, the choice the perturbation series makes. Where the series converges at $\lambda = 1$ its sum equals this result; where it does not, the diagonalisation is its analytic continuation, the unique function whose Taylor series it is.

Comparison with first-order CHP

	First order	All orders
S_{pq}	$(pq\|cc)/(\epsilon_q - \epsilon_p)$, antisymmetric	U_{qp}, not antisymmetric
S_{pp}	set to 1	< 1 (normalised)
Hole level shift $-(pp\|cc)$	absent	included
Valid when	$\|V_{pq}\|,\ \|V_{pp} - V_{qq}\| \ll \|\epsilon_p - \epsilon_q\|$	always defined

For bromothiophene (singlet, core ψ_c = mo:31, valence RAS2 block), $|S_{pq}|$:

	First order	All orders	Overlaps method
$S_{34,38}$ / $S_{38,34}$	0.365 / 0.365	0.190 / 0.096	0.127 / 0.126
$S_{38,40}$ / $S_{40,38}$	0.267 / 0.267	0.150 / 0.194	0.057 / 0.153
$S_{34,40}$ / $S_{40,34}$	0.031 / 0.031	0.010 / 0.032	0.006 / 0.022
$S_{38,38}$	1	0.595	0.712

For ϕ_{34} and ϕ_{38} the series has a radius of convergence of $|\lambda| \approx 0.37$ (from the growth of $c^{(k)}$; partial sums converge to the diagonalisation at $\lambda = 0.1$ and 0.3 and oscillate at $\lambda = 0.5$ and 1). The hole lowers ψ_{38} by about $0.4\,E_h$, more than its gaps to its neighbours, so no finite order converges at the physical strength. The first-order derivative $\partial S_{pq}/\partial\lambda|_{\lambda=0}$ of the diagonalisation reproduces Eq. (1) exactly.

Assumptions and limitations

- **The initial orbitals are assumed to diagonalise $\hat{f}_0$.** The SA natural orbitals diagonalise the state-averaged density, not the generalised Fock matrix $\mathbf{F} = \mathbf{F}^I + \mathbf{F}^A$; $\epsilon_p = F_{pp}$, but $F_{pq} \neq 0$ (up to $0.013\,E_h$ in the valence block, against $(pq|cc)$ up to $0.15\,E_h$). Dropping F_{pq} treats the natural orbitals as canonical. Using the full $\mathbf{F} - \mathbf{J}_c$ instead would attribute the mixing caused by $\mathbf{F}$ itself to the hole ($S \neq 1$ at $\mathbf{J}_c = 0$), unless the initial orbitals are redefined as eigenvectors of $\mathbf{F}$, which are no longer the orbitals the CI vectors are expressed in. The effect on S is 0.01–0.04 here.
- **State averaging.** ϵ_p are diagonal elements of a state-averaged Fock matrix, not orbital energies of a particular state, and their order is not guaranteed.
- **Frozen, non-self-consistent potential.** $\hat{V}$ is the bare Coulomb field of the removed electron in the initial ψ_c; the response of the other electrons (screening) and the relaxation of ψ_c are not iterated.
- **Coulomb only.** The spin-averaged change of the Fock operator is $-\hat{J}_c + \frac{1}{2}\hat{K}_c$; the exchange term is omitted (it changes S little), so S is the same for every final spin.
- **Truncated space.** Only the valence active orbitals are coupled; mixing with orbitals outside the active space is neglected.
- **Near-degeneracy.** When the level shift brings two levels together (ψ_{38} next to ψ_{39}), the relaxed orbitals are strongly mixed ($S_{38,38} \approx 0.6$), the maximum-overlap labels and phases become ambiguous, and the result depends on initial-state gaps that the hole has invalidated.